

DAT 260 A Comparison of SQL and No-SQL Database Management Systems for Unstructured Data Text-Only Versions

Figure 1: SQL Breakups

A flowchart is displayed. It features five text boxes. A text box titled SQL Language Statements is displayed which connects to the following text boxes:

- DML: SELECT, INSERT, UPDATE, DELETE.
- DDL: CREATE, ALTER, DROP
- DCL: GRANT, REVOKE
- TCL: BEGIN, TRAIN, COMMIT, TRAN, ROLLBACK

Figure 2: Properties of NoSQL databases

An infographic is displayed. It features three downward arrows named as 1, 2, and 3. Each Arrow features two bullet points as follows:

- Arrow 1
 - Not schema based database
 - Supports many data types at a time
- Arrow 2
 - No referential key, easy to handle
 - Flexible
- Arrow 3
 - Non-relational database
 - Supports JSON format

Figure 3: SQL Vs MongoDB terms

A diagram is displayed. There are two text boxes titled SQL Terms slash Concepts and MongoDB Terms slash Concepts. The following text is displayed under SQL Terms slash Concepts:

- database
- tables
- rows
- columns

The following text is displayed under MongoDB Terms/Concepts:

- database
- collections
- documents (BSON)
- fields

The following elements under SQL Terms slash Concepts and MongoDB Terms slash Concepts are connected with dotted lines:

- database and database
- tables and collections
- rows and documents (BSON)
- columns and fields

Figure 4: Result of Experiment 1

A bar graph is displayed. The x-axis represents numbers from 1 to 9. The y-axis represents numbers from 0 to 180, at intervals of 20. The following legend is displayed:

- Shade 1: SQL
- Shade 2: Mongo

All the points on the y-axis have two bars paired together representing SQL and Mongo. The details of the graph are as follows:

- 1

- SQL- 159
 - Mongo- 86
- 2
 - SQL- 122
 - Mongo- 109
- 3
 - SQL- 153
 - Mongo- 129
- 4
 - SQL- 170
 - Mongo- 130
- 5
 - SQL- 150
 - Mongo- 134
- 6
 - SQL- 148
 - Mongo- 138
- 7
 - SQL- 155
 - Mongo- 110
- 8
 - SQL-144
 - Mongo- 127
- 9
 - SQL- 140
 - Mongo- 121

Figure 5: Result of Experiment 2.

A bar graph is displayed. The x-axis represents numbers from 1 to 9. The y-axis represents numbers from 0 to 350, at intervals of 50. The following legend is displayed:

- Shade 1: SQL
- Shade 2: Mongo

All the points on the y-axis have two bars paired together representing SQL and Mongo. The details of the graph are as follows:

- 1
 - SQL- 209
 - Mongo- 142
- 2
 - SQL- 207
 - Mongo- 123
- 3
 - SQL- 279
 - Mongo- 104
- 4
 - SQL- 287
 - Mongo- 112
- 5
 - SQL- 285
 - Mongo- 123
- 6
 - SQL- 211

- 7
 - Mongo- 147
 - SQL- 257
 - Mongo- 115
- 8
 - SQL- 208
 - Mongo- 113
- 9
 - SQL- 227
 - Mongo- 134